

Part B Question 4

Algorithm :- Take input all the values and check if they are valid or not. Calculate the final bill by applying the all three functions. Print the final bill.

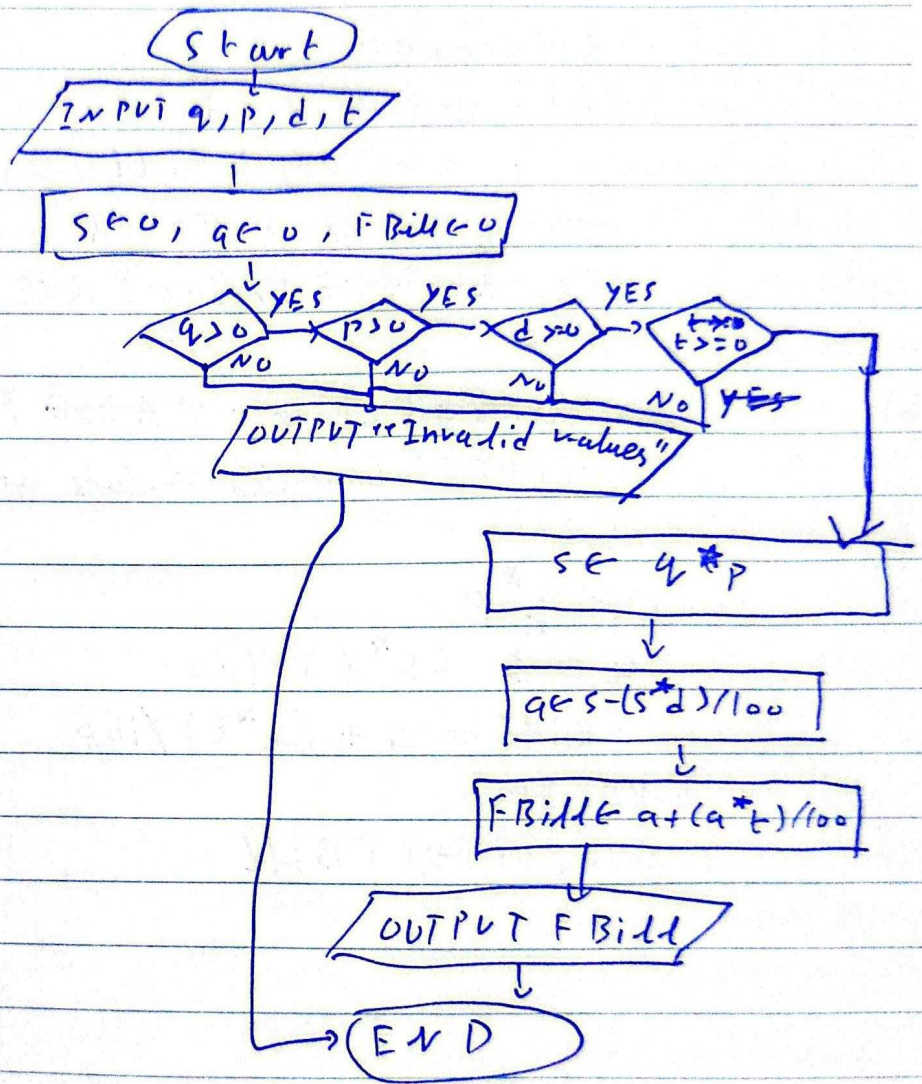
PAC :-

Given Data	Process	Required Results
• Quantity • Price • Discount • Tax	• Check if all the values are valid • Calculate Subtotal • Calculate Discounted amount • Calculate Final Bill	• Final Bill

IPO Chart :-

Input	Process	OUTPUTS
• q • p • d • t	• Check if all values are valid • $s = q * p$ • $a = s - (s * d) / 100$ • $F \text{ Bill} = a + (a * t) / 100$	• F Bill

Flow Chart:-



Pseudocode:

INPUT q, p, d, t

$s \leftarrow 0, a \leftarrow 0, FBill \leftarrow 0, Error \leftarrow FALSE$

IF ~~$q < 0$~~ $q \leq 0$ OR $p < 0$ OR $d < 0$, OR $t < 0$ THEN
~~ERROR TRUE~~ $Error \leftarrow TRUE$

ENDIF

IF ~~ERROR~~ $Error = FALSE$ THEN
 OUTPUT "Invalid v-values are entered"

ELSE

$s \leftarrow q * p$

$a \leftarrow s - (s * d) / 100$

$FBill \leftarrow a + (a * t) / 100$

ENDIF

OUTPUT ~~FBill~~